

# A Comparative Analysis of Income Level Prediction

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# Data

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The dataset is from 1994; census income

<https://archive.ics.uci.edu/ml/datasets/Census+Income>

14 attributes

32561 records.

| age                    |        | workclass         |        | fnlwgt          |          | education           |        | educationnum   |        |             |        |
|------------------------|--------|-------------------|--------|-----------------|----------|---------------------|--------|----------------|--------|-------------|--------|
| Min.                   | :17.00 | Private           | :22696 | Min.            | : 12285  | HS-grad             | :10501 | Min.           | : 1.00 |             |        |
| 1st Qu.:               | 28.00  | Self-emp-not-inc: | 2541   | 1st Qu.:        | 117827   | Some-college:       | 7291   | 1st Qu.:       | 9.00   |             |        |
| Median                 | :37.00 | Local-gov         | : 2093 | Median          | : 178356 | Bachelors           | : 5355 | Median         | :10.00 |             |        |
| Mean                   | :38.58 | State-gov         | : 1298 | Mean            | : 189778 | Masters             | : 1723 | Mean           | :10.08 |             |        |
| 3rd Qu.:               | 48.00  | Self-emp-inc      | : 1116 | 3rd Qu.:        | 237051   | Assoc-voc           | : 1382 | 3rd Qu.:       | 12.00  |             |        |
| Max.                   | :90.00 | (Other)           | : 981  | Max.            | :1484705 | 11th                | : 1175 | Max.           | :16.00 |             |        |
|                        |        | NA's              | : 1836 |                 |          | (Other)             | : 5134 |                |        |             |        |
| maritalstatus          |        | occupation        |        | relationship    |          | race                |        |                |        |             |        |
| Divorced               | : 4443 | Prof-specialty    | : 4140 | Husband         | :13193   | Amer-Indian-Eskimo: | 311    |                |        |             |        |
| Married-AF-spouse      | : 23   | Craft-repair      | : 4099 | Not-in-family   | : 8305   | Asian-Pac-Islander: | 1039   |                |        |             |        |
| Married-civ-spouse     | :14976 | Exec-managerial:  | 4066   | Other-relative: | 981      | Black               | : 3124 |                |        |             |        |
| Married-spouse-absent: | 418    | Adm-clerical      | : 3770 | Own-child       | : 5068   | Other               | : 271  |                |        |             |        |
| Never-married          | :10683 | Sales             | : 3650 | Unmarried       | : 3446   | White               | :27816 |                |        |             |        |
| Separated              | : 1025 | (Other)           | :10993 | Wife            | : 1568   |                     |        |                |        |             |        |
| Widowed                | : 993  | NA's              | : 1843 |                 |          |                     |        |                |        |             |        |
| sex                    |        | capitalgain       |        | capitalloss     |          | hoursperweek        |        | nativecountry  |        | incomelevel |        |
| Female:                | 10771  | Min.              | : 0    | Min.            | : 0.0    | Min.                | : 1.00 | United-States: | 29170  | <=50K:      | 24720  |
| Male                   | :21790 | 1st Qu.:          | 0      | 1st Qu.:        | 0.0      | 1st Qu.:            | 40.00  | Mexico         | : 643  | >50K        | : 7841 |
|                        |        | Median            | : 0    | Median          | : 0.0    | Median              | :40.00 | Philippines    | : 198  |             |        |
|                        |        | Mean              | : 1078 | Mean            | : 87.3   | Mean                | :40.44 | Germany        | : 137  |             |        |
|                        |        | 3rd Qu.:          | 0      | 3rd Qu.:        | 0.0      | 3rd Qu.:            | 45.00  | Canada         | : 121  |             |        |
|                        |        | Max.              | :99999 | Max.            | :4356.0  | Max.                | :99.00 | (Other)        | : 1709 |             |        |
|                        |        |                   |        |                 |          |                     |        | NA's           | : 583  |             |        |



# NA Values

First attempt was assigning average weights, however, later in the random forest analysis, the NA's occurred again, so my second approach was to remove all NA values.

```
inc_clean <- DMwR::knnImputation(inc, k=5, scale=T, meth="weighAve")
```

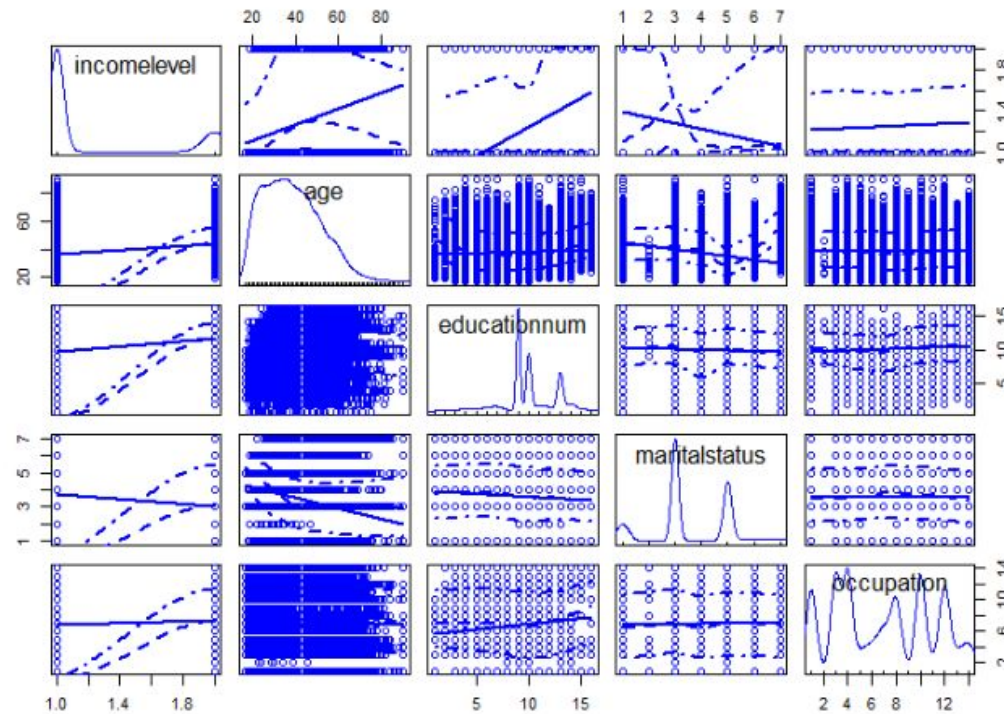
```
summary(inc_clean)
```

```
inc_clean <- inc[complete.cases(inc), ]
```

```
> summary(inc_clean)
```

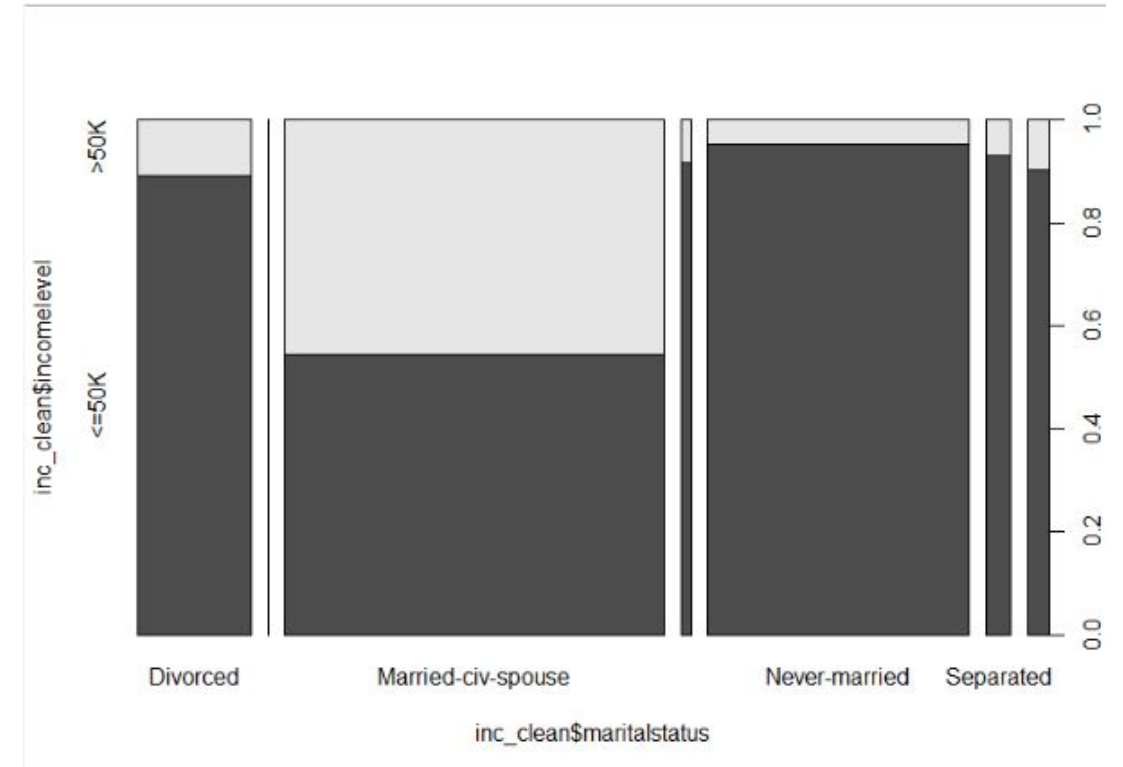
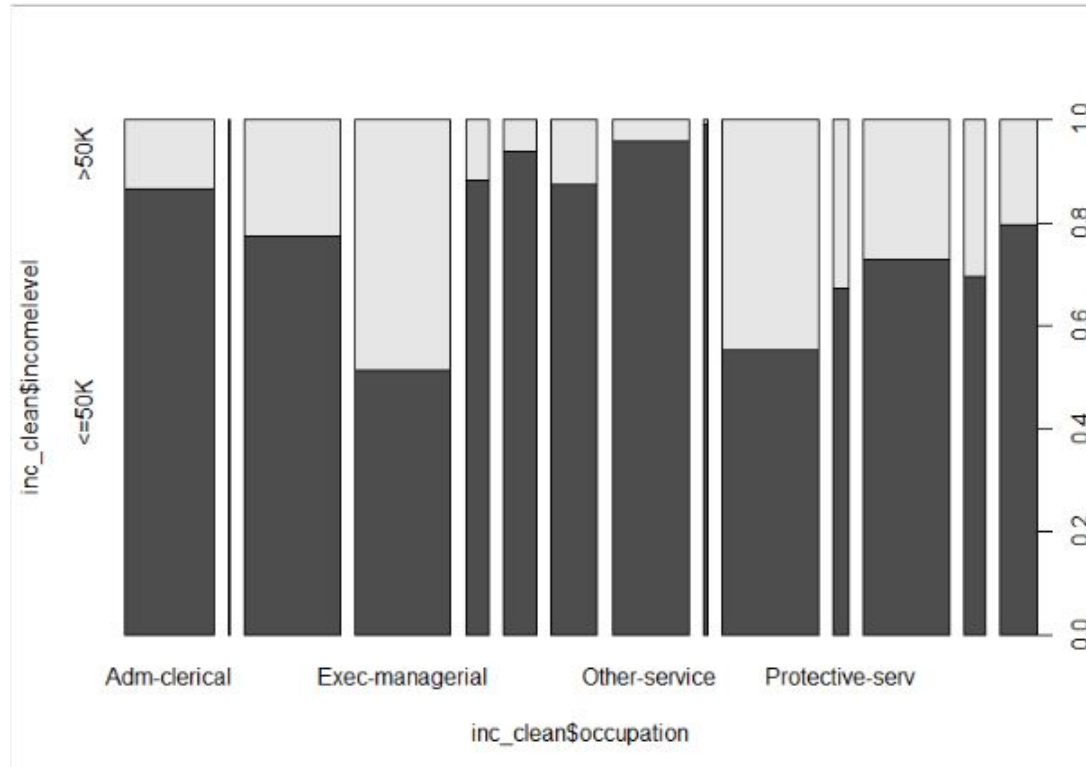
| age                   |        | workclass        |        | fnlwgt         |          | education          |        | educationnum  |        |             |       |
|-----------------------|--------|------------------|--------|----------------|----------|--------------------|--------|---------------|--------|-------------|-------|
| Min.                  | :17.00 | Private          | :22286 | Min.           | : 13769  | HS-grad            | :9840  | Min.          | : 1.00 |             |       |
| 1st Qu.:              | 28.00  | Self-emp-not-inc | : 2499 | 1st Qu.:       | 117627   | Some-college       | :6678  | 1st Qu.:      | 9.00   |             |       |
| Median                | :37.00 | Local-gov        | : 2067 | Median         | : 178425 | Bachelors          | :5044  | Median        | :10.00 |             |       |
| Mean                  | :38.44 | State-gov        | : 1279 | Mean           | : 189794 | Masters            | :1627  | Mean          | :10.12 |             |       |
| 3rd Qu.:              | 47.00  | Self-emp-inc     | : 1074 | 3rd Qu.:       | 237629   | Assoc-voc          | :1307  | 3rd Qu.:      | 13.00  |             |       |
| Max.                  | :90.00 | Federal-gov      | : 943  | Max.           | :1484705 | 11th               | :1048  | Max.          | :16.00 |             |       |
|                       |        | (Other)          | : 14   |                |          | (Other)            | :4618  |               |        |             |       |
| maritalstatus         |        | occupation       |        | relationship   |          | race               |        |               |        |             |       |
| Divorced              | : 4214 | Prof-specialty   | :4038  | Husband        | :12463   | Amer-Indian-Eskimo | : 286  |               |        |             |       |
| Married-AF-spouse     | : 21   | Craft-repair     | :4030  | Not-in-family  | : 7726   | Asian-Pac-Islander | : 895  |               |        |             |       |
| Married-civ-spouse    | :14065 | Exec-managerial  | :3992  | Other-relative | : 889    | Black              | : 2817 |               |        |             |       |
| Married-spouse-absent | : 370  | Adm-clerical     | :3721  | Own-child      | : 4466   | Other              | : 231  |               |        |             |       |
| Never-married         | : 9726 | Sales            | :3584  | Unmarried      | : 3212   | White              | :25933 |               |        |             |       |
| Separated             | : 939  | Other-service    | :3212  | wife           | : 1406   |                    |        |               |        |             |       |
| Widowed               | : 827  | (Other)          | :7585  |                |          |                    |        |               |        |             |       |
| sex                   |        | capitalgain      |        | capitalloss    |          | hoursperweek       |        | nativecountry |        | incomelevel |       |
| Female:               | 9782   | Min.             | : 0    | Min.           | : 0.00   | Min.               | : 1.00 | United-States | :27504 | <=50K:      | 22654 |
| Male                  | :20380 | 1st Qu.:         | 0      | 1st Qu.:       | 0.00     | 1st Qu.:           | 40.00  | Mexico        | : 610  | >50K:       | 7508  |
|                       |        | Median           | : 0    | Median         | : 0.00   | Median             | :40.00 | Philippines   | : 188  |             |       |
|                       |        | Mean             | : 1092 | Mean           | : 88.37  | Mean               | :40.93 | Germany       | : 128  |             |       |
|                       |        | 3rd Qu.:         | 0      | 3rd Qu.:       | 0.00     | 3rd Qu.:           | 45.00  | Puerto-Rico   | : 109  |             |       |
|                       |        | Max.             | :99999 | Max.           | :4356.00 | Max.               | :99.00 | Canada        | : 107  |             |       |
|                       |        |                  |        |                |          |                    |        | (Other)       | : 1516 |             |       |

# Data Exploration



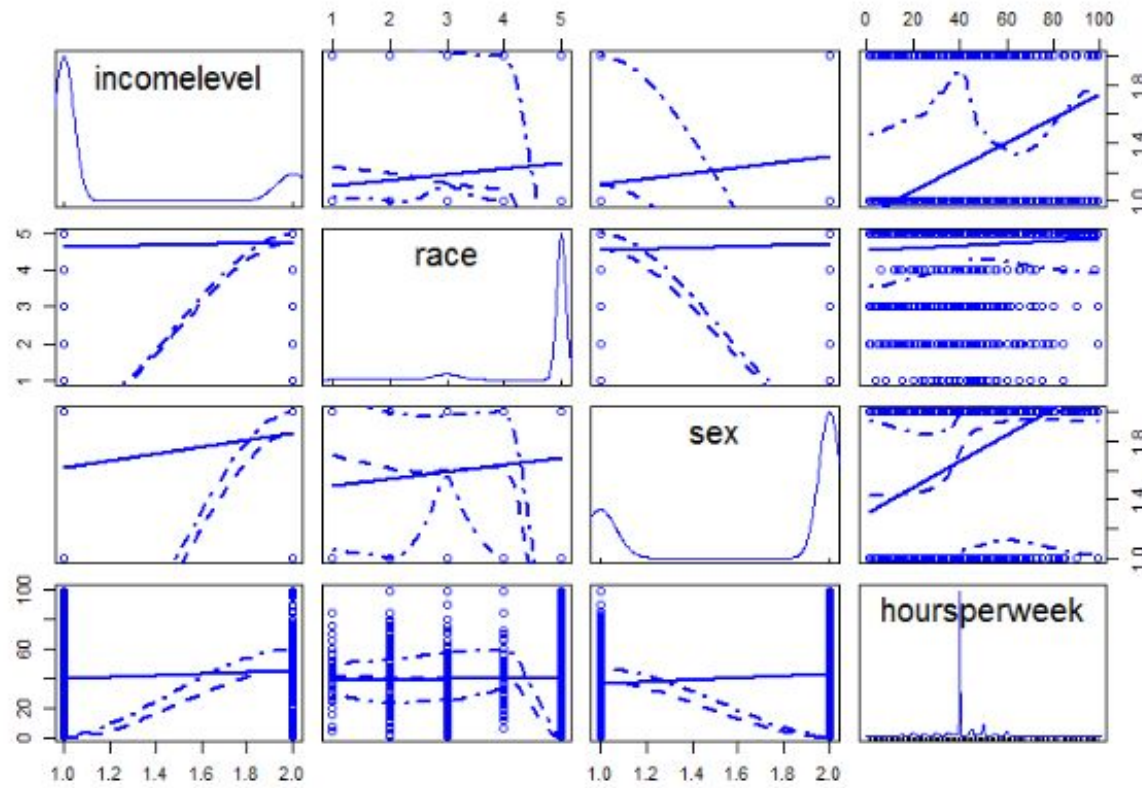
Income level vs. age, education num,  
marital status; married people earns  
higher than single or widowed.  
, occupation

# Income Level vs. Occupation, Marital Status

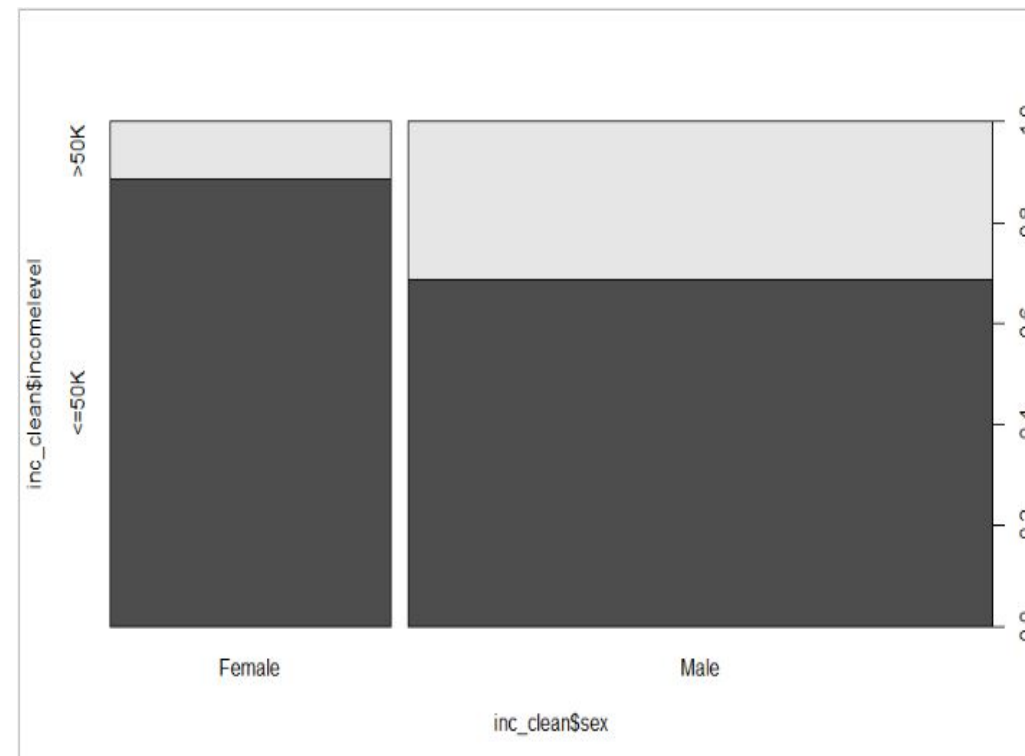
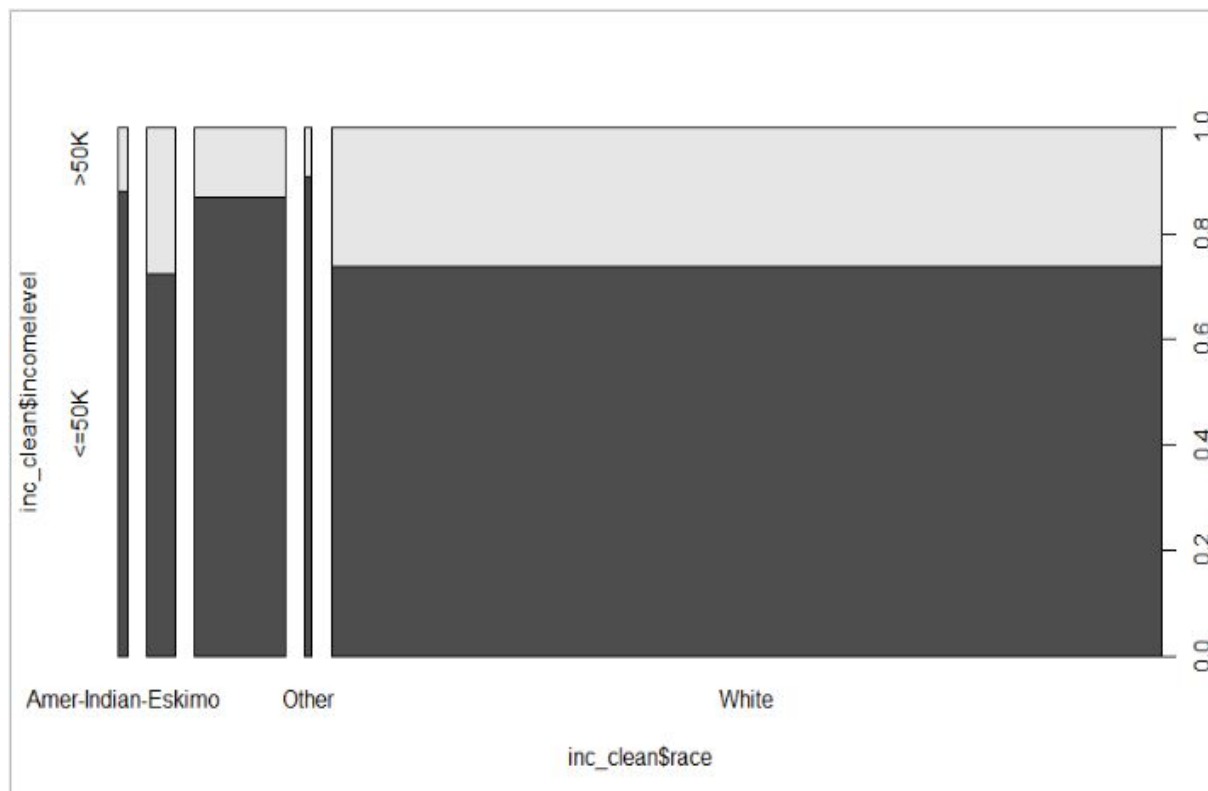


# Data Exploration

---



# Income Level vs. Race, Gender



# Divide dataset into Training and Test

---

```
set.seed(12345)

# split training and test datasets, 80% and 20%

train_sample <- sample(nrow(inc_clean), round(nrow(inc_clean)*0.8))

inc_clean_train <- inc[train_sample, ]

inc_clean_test  <- inc[-train_sample, ]
```

# Logistic Regression Model

---

```
log.base.model <- glm(incomelevel~., data=inc_clean_train,  
family="binomial")
```

- `vif(log.base.model)` Error in `vif.default(log.base.model)` : there are aliased coefficients in the model
- Education and educationnum are representing the same information, one with categorical and the latter is with numeric. So I removed the categorical education

# Logistic Regression Model 2

---

```
inc_clean <- inc_clean[,-4]

inc_clean_train <- inc_clean_train[,-4]

inc_clean_test <-
inc_clean_test[,-4]log.updated.model2 <-
glm(incomelevel~age+workclass+fnlwgt+educatio
nnum+
occupation+relationship+race+sex+capitalgain+
capitalloss+
hoursperweek, data=inc_clean_train,
family="binomial")
```

|               | GVIF       | Df | GVIF^(1/(2*Df)) |
|---------------|------------|----|-----------------|
| age           | 1.201917   | 1  | 1.096320        |
| workclass     | 1.554361   | 6  | 1.037439        |
| fnlwgt        | 1.052566   | 1  | 1.025946        |
| educationnum  | 1.495792   | 1  | 1.223026        |
| maritalstatus | 57.379362  | 6  | 1.401403        |
| occupation    | 2.492011   | 13 | 1.035743        |
| relationship  | 131.260884 | 5  | 1.628597        |
| race          | 3.279240   | 4  | 1.160037        |
| sex           | 2.939508   | 1  | 1.714499        |
| capitalgain   | 1.034619   | 1  | 1.017162        |
| capitalloss   | 1.014217   | 1  | 1.007084        |
| hoursperweek  | 1.130079   | 1  | 1.063052        |
| nativecountry | 3.545161   | 40 | 1.015946        |

# Final Logistic Regression model

```
Call:
glm(formula = incomelevel ~ age + workclass + fnlwgt + educationnum +
    occupation + relationship + race + sex + capitalgain + capitalloss +
    hoursperweek, family = "binomial", data = inc_clean_train)
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q      | Max    |
|---------|---------|---------|---------|--------|
| -4.9840 | -0.5202 | -0.1962 | -0.0185 | 3.3671 |

Coefficients:

|                             | Estimate   | Std. Error | z value | Pr(> z )     |
|-----------------------------|------------|------------|---------|--------------|
| (Intercept)                 | -6.949e+00 | 3.381e-01  | -20.555 | < 2e-16 ***  |
| age                         | 2.777e-02  | 1.879e-03  | 14.779  | < 2e-16 ***  |
| workclassLocal-gov          | -8.485e-01 | 1.311e-01  | -6.470  | 9.81e-11 *** |
| workclassPrivate            | -5.146e-01 | 1.084e-01  | -4.746  | 2.08e-06 *** |
| workclassSelf-emp-inc       | -3.439e-01 | 1.427e-01  | -2.410  | 0.015967 *   |
| workclassSelf-emp-not-inc   | -1.028e+00 | 1.271e-01  | -8.087  | 6.12e-16 *** |
| workclassState-gov          | -7.989e-01 | 1.440e-01  | -5.548  | 2.89e-08 *** |
| workclasswithout-pay        | -1.219e+01 | 1.306e+02  | -0.093  | 0.925637     |
| fnlwgt                      | 6.838e-07  | 2.012e-07  | 3.399   | 0.000675 *** |
| educationnum                | 2.708e-01  | 1.099e-02  | 24.633  | < 2e-16 ***  |
| occupationArmed-Forces      | -8.075e-01 | 1.672e+00  | -0.483  | 0.629213     |
| occupationCraft-repair      | 2.536e-02  | 9.270e-02  | 0.274   | 0.784432     |
| occupationExec-managerial   | 8.225e-01  | 8.882e-02  | 9.260   | < 2e-16 ***  |
| occupationFarming-fishing   | -1.025e+00 | 1.646e-01  | -6.226  | 4.80e-10 *** |
| occupationHandlers-cleaners | -6.817e-01 | 1.644e-01  | -4.145  | 3.39e-05 *** |
| occupationMachine-op-inspct | -3.400e-01 | 1.179e-01  | -2.884  | 0.003926 **  |
| occupationOther-service     | -8.771e-01 | 1.366e-01  | -6.421  | 1.35e-10 *** |
| occupationPriv-house-serv   | -3.357e+00 | 1.861e+00  | -1.804  | 0.071171 .   |
| occupationProf-specialty    | 6.247e-01  | 9.215e-02  | 6.780   | 1.20e-11 *** |
| occupationProtective-serv   | 5.143e-01  | 1.473e-01  | 3.492   | 0.000479 *** |
| occupationSales             | 2.325e-01  | 9.582e-02  | 2.426   | 0.015265 *   |
| occupationTech-support      | 6.358e-01  | 1.262e-01  | 5.039   | 4.67e-07 *** |
| occupationTransport-moving  | -1.224e-01 | 1.152e-01  | -1.063  | 0.287874     |

|                            |            |           |         |             |
|----------------------------|------------|-----------|---------|-------------|
| relationshipNot-in-family  | -1.851e+00 | 6.664e-02 | -27.781 | < 2e-16 *** |
| relationshipOther-relative | -2.293e+00 | 2.577e-01 | -8.899  | < 2e-16 *** |
| relationshipOwn-child      | -3.080e+00 | 1.730e-01 | -17.805 | < 2e-16 *** |
| relationshipUnmarried      | -1.782e+00 | 1.145e-01 | -15.571 | < 2e-16 *** |
| relationshipWife           | 1.405e+00  | 1.202e-01 | 11.695  | < 2e-16 *** |
| raceAsian-Pac-Islander     | 4.715e-01  | 2.808e-01 | 1.679   | 0.093104 .  |
| raceBlack                  | 3.493e-01  | 2.671e-01 | 1.308   | 0.191014    |
| raceOther                  | -2.322e-01 | 4.329e-01 | -0.537  | 0.591585    |
| raceWhite                  | 4.737e-01  | 2.539e-01 | 1.866   | 0.062088 .  |
| sexMale                    | 8.659e-01  | 9.173e-02 | 9.439   | < 2e-16 *** |
| capitalgain                | 3.090e-04  | 1.199e-05 | 25.762  | < 2e-16 *** |
| capitalloss                | 6.176e-04  | 4.567e-05 | 13.525  | < 2e-16 *** |
| hoursperweek               | 3.157e-02  | 1.977e-03 | 15.964  | < 2e-16 *** |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 25060 on 22371 degrees of freedom  
Residual deviance: 14569 on 22336 degrees of freedom  
AIC: 14641

Number of Fisher Scoring iterations: 12

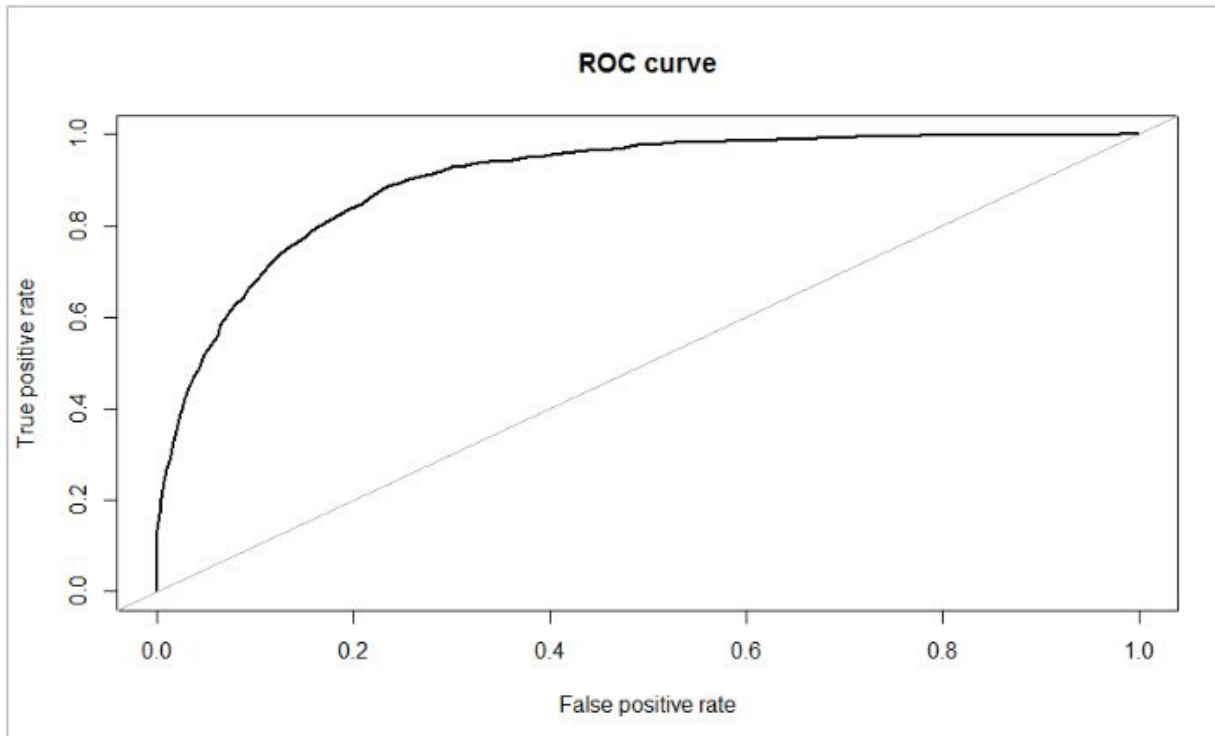
# Final logistic regression model VIF

---

```
> vif(log.updated.model2)
```

|              | GVIF     | Df | GVIF^(1/(2*Df)) |
|--------------|----------|----|-----------------|
| age          | 1.125789 | 1  | 1.061032        |
| workclass    | 1.517796 | 6  | 1.035383        |
| fnlwgt       | 1.034544 | 1  | 1.017126        |
| educationnum | 1.476600 | 1  | 1.215154        |
| occupation   | 2.365472 | 13 | 1.033669        |
| relationship | 3.147521 | 5  | 1.121494        |
| race         | 1.078138 | 4  | 1.009449        |
| sex          | 2.827545 | 1  | 1.681530        |
| capitalgain  | 1.028774 | 1  | 1.014285        |
| capitalloss  | 1.009528 | 1  | 1.004753        |
| hoursperweek | 1.122055 | 1  | 1.059271        |

# Logistic Regression model Evaluation



| actual default | predicted default |               | Row Total     |
|----------------|-------------------|---------------|---------------|
|                | <=50K             | >50K          |               |
| <=50K          | 5460<br>0.935     | 377<br>0.065  | 5837<br>0.749 |
| >50K           | 818<br>0.419      | 1133<br>0.581 | 1951<br>0.251 |
| column Total   | 6278              | 1510          | 7788          |

```
> library(ROSE)
Loaded ROSE 0.0-3

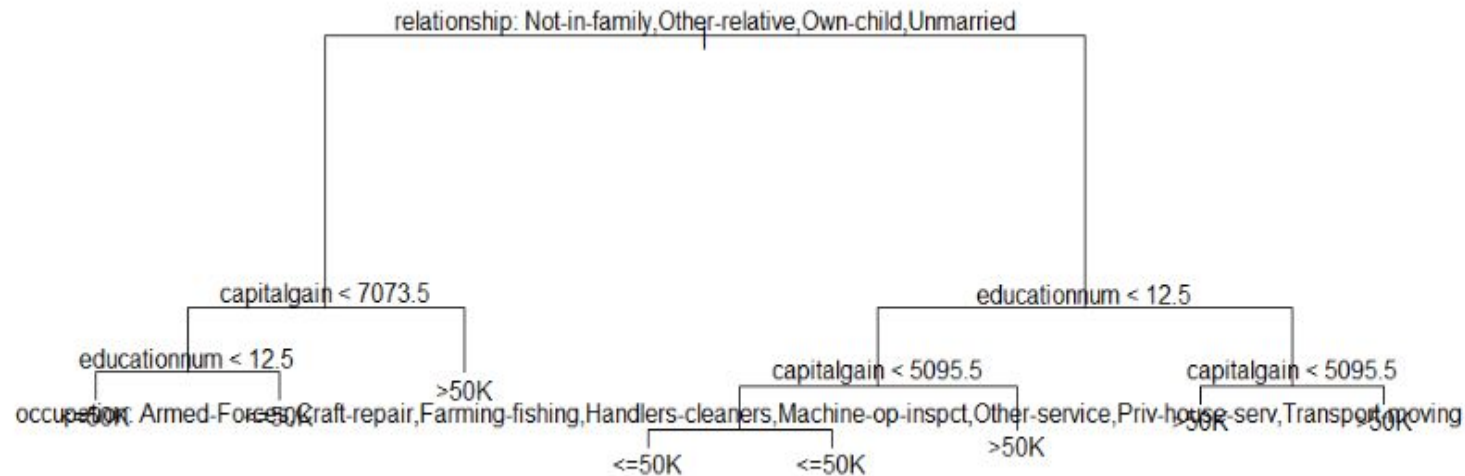
> roc.curve(inc_clean_test$incomelevel, predictLogit)
Area under the curve (AUC): 0.905
```

**Misclassification =>  $818 + 377 / 7788 = 0.153$**

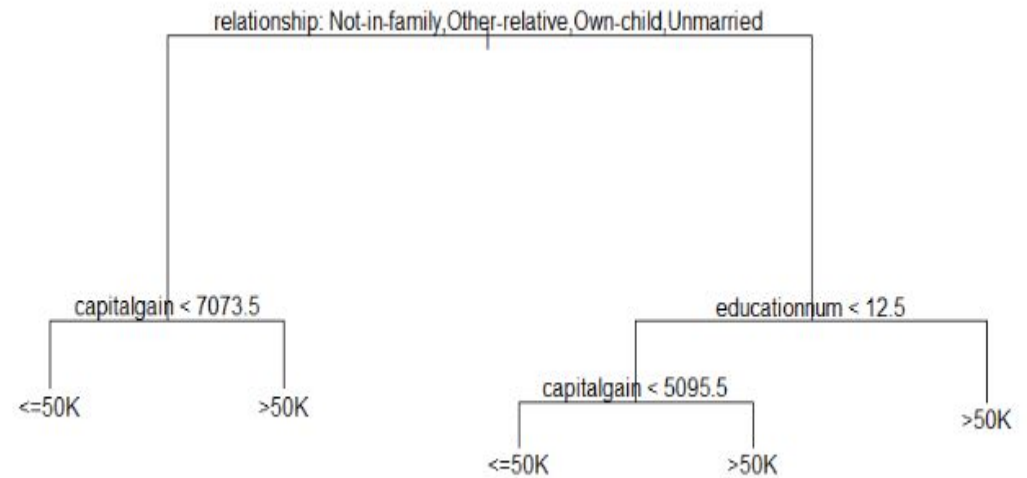
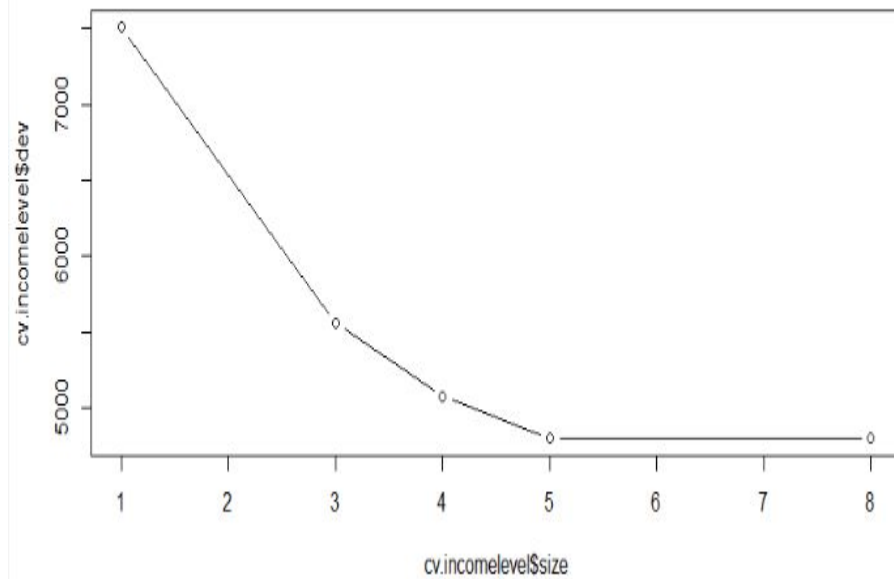
# Tree based Classification

---

```
tree.incomelevel <- tree(incomelevel~., data=inc_clean_train)
Error in tree(incomelevel ~ ., data = inc_clean_train) :
factor predictors must have at most 32 levels
```



# Pruning Tree



# Tree Evaluation

---

| actual default | predicted default |              | Row Total     |
|----------------|-------------------|--------------|---------------|
|                | <=50K             | >50K         |               |
| <=50K          | 5546<br>0.950     | 291<br>0.050 | 5837<br>0.749 |
| >50K           | 964<br>0.494      | 987<br>0.506 | 1951<br>0.251 |
| Column Total   | 6510              | 1278         | 7788          |

**Misclassification =>  $291+964/7788 = 0.1611$**

# Random Forest Classification

---

```
> rf.incomelevel <- randomForest(incomelevel~., data=inc_clean, importance=TRUE)
> importance(rf.incomelevel)
```

|               | <=50K      | >50K         | MeanDecreaseAccuracy | MeanDecreaseGini |
|---------------|------------|--------------|----------------------|------------------|
| age           | 3.641301   | 100.76188909 | 84.44326             | 1123.9928        |
| workclass     | 39.772265  | 13.04294709  | 41.90885             | 356.5142         |
| fnlwgt        | 2.585738   | 2.10524763   | 3.46180              | 1088.2862        |
| educationnum  | 54.916706  | 106.31182989 | 117.07856            | 1058.5346        |
| maritalstatus | 55.469090  | 23.12517560  | 44.79723             | 972.0021         |
| occupation    | 47.897830  | 75.38248904  | 103.21965            | 980.7648         |
| relationship  | 27.022491  | 37.88093967  | 35.72363             | 1129.6864        |
| race          | 12.660311  | 8.54214761   | 16.61481             | 126.3652         |
| sex           | 31.051245  | 5.43487346   | 35.93561             | 136.4420         |
| capitalgain   | 140.001265 | 214.66007904 | 192.39871            | 1208.5744        |
| capitalloss   | 64.179148  | 101.32815497 | 98.30473             | 357.9737         |
| hoursperweek  | 9.519741   | 64.49508414  | 52.90756             | 669.8446         |
| nativecountry | 24.862517  | -0.03871897  | 20.97168             | 244.5563         |

```
< |
```

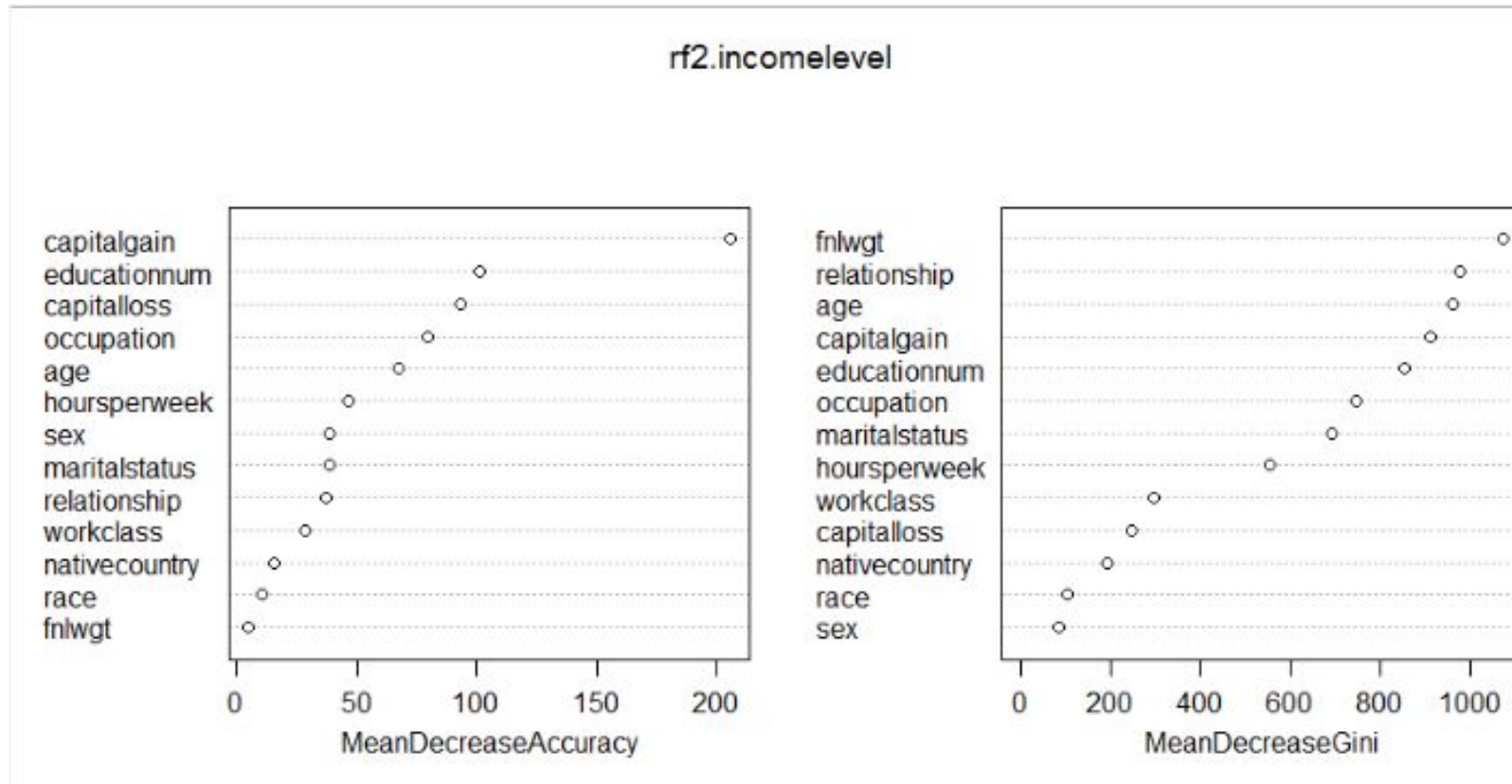
The gini index shows that age, education, and the relationship are all important variables.

# Random Forest Tuning

---

```
##### Tune Random Forest #####  
> mtry <- tuneRF(inc_clean_train, inc_clean_train$incomelevel, ntreeTry=10,  
+ stepFactor=1.5, improve=TRUE, trace=TRUE, plot=TRUE, doBest=TRUE)  
mtry = 3 OOB error = 0.16% Searching left ... mtry = 2 OOB error = 0.52% -2.190979  
TRUE Searching right ... mtry = 4 OOB error = 0.09% 0.4730805 TRUE  
  
rf2.incomelevel <- randomForest(incomelevel~., data=inc_clean_train, mtry=4,  
importance=TRUE) > varImpPlot(rf2.incomelevel)
```

# Random forest Variable Importance Plot



# Random Forest Evaluation

---

| actual default | predicted default |               | Row Total     |
|----------------|-------------------|---------------|---------------|
|                | <=50K             | >50K          |               |
| <=50K          | 5398<br>0.925     | 439<br>0.075  | 5837<br>0.749 |
| >50K           | 692<br>0.355      | 1259<br>0.645 | 1951<br>0.251 |
| Column Total   | 6090              | 1698          | 7788          |

**Misclassification** =>  $439+692/7788 = 0.1452$

# SVM Classification, Linear

---

```
svm.fit <- svm(incomelevel~.,data=inc_clean_train, cost=0.01, kernel='linear')
```

```
summary(svm.fit)
```

# SVM Evaluation, Linear

---

| actual default | predicted default |               | Row Total     |
|----------------|-------------------|---------------|---------------|
|                | <=50K             | >50K          |               |
| <=50K          | 5460<br>0.935     | 377<br>0.065  | 5837<br>0.749 |
| >50K           | 818<br>0.419      | 1133<br>0.581 | 1951<br>0.251 |
| Column Total   | 6278              | 1510          | 7788          |

**Misclassification =>  $818+377/7788 = 0.1534$**

# Comparison

---

| Logistic Regression     | <b>Misclassification =&gt; <math>818+377/7788 = 0.153</math></b>  |
|-------------------------|-------------------------------------------------------------------|
| Pruned Tree             | <b>Misclassification =&gt; <math>291+964/7788 = 0.1611</math></b> |
| Optimized Random Forest | <b>Misclassification =&gt; <math>439+692/7788 = 0.1452</math></b> |
| Linear SVM              | <b>Misclassification =&gt; <math>818+377/7788 = 0.1534</math></b> |

# Discussion and Conclusion

---

Fine tuning of SVM could perform better (**both for linear and polynomial**).

The tuned random forest model has the lowest classification rate (**optimized mtry variable**).

Logistic regression model is strong in predicting the income level.